PROBABILITY AND STATISTICS FOR NON-PATTERN PROBLEM

CHAPTER 6

- SAMPLING DISTRIBUTION AND POINT ESTIMATION OF PARAMETERS

CHAPTER OUTLINE

Sampling Distributions

- Definition
- Sampling Distributions and the central limit theorem
- Point Estimation
- Interval Estimation

SAMPLING DISTRIBUTION

- There are 3 distinct types of distribution of data:
  - **Population Distribution**, characterizes the distribution elements of a population
  - **Sample Distribution,** characterizes the distribution elements of sample drawn from a population
  - **Sampling Distribution,** A probability distribution of a **statistic** that comes from choosing random samples of a given population.

- “Statistics" generally refers to **numerical characteristics** that are calculated from a sample of data.
- These statistics provide insights into the properties of the underlying population from which the sample was drawn.

DEFINITION

Sampling distributions constitute the theoretical basis of statistical inference.

Sampling distribution provide a major simplification to statistical inference.

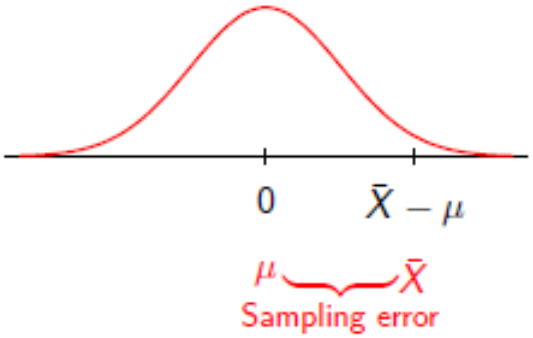
- A **sampling distribution** is a theoretical probability distribution of a statistic obtained through a large number of samples drawn from a specific population.

SAMPLING ERROR, VARIATION AND DISTRIBUTION

- Sampling error = $\bar{X} - \mu$ is unknown, <u>cannot</u> be estimated
- Sampling variation is variation in $\bar{X}$

  Variation in $\bar{X}$ = variation in $(\bar{X} - \mu)$

  … since μ is constant



- Sampling variation <u>can be</u> estimated
- **Normal distribution** was used as a distribution for measurement errors. Therefore, we conjecture the sampling error distribution to be a normal distribution.
- The sampling error distribution is called a **sampling distribution**

EXAMPLE : SAMPLING VARIATION

		X	$=$	$\bar{X}$	
n = 1	Sample 1	1874		**1874**	Large difference
	Sample 2	1514		**1514**	

		$\frac{1}{2}(X_1 + X_2)$		$=$	$\bar{X}$	
n = 2	Sample 1	1874	1642		**1758**	smaller difference
	Sample 2	1514	2358		**1936**	

		$\frac{1}{2}(X_1 + X_2 + \cdots + X_{10})$				$=$	$\bar{X}$	
n = 10	Sample 1	1874	1642	…	1918		**1708**	Smallest difference
	Sample 2	1514	2358	…	1809		**1914**	

TYPE OF SAMPLING DISTRIBUTION

- There are two types of sampling distribution
  - **Sampling Distribution of the Mean**
  - **Sampling Distribution of the Proportion**

TYPE OF SAMPLING DISTRIBUTION

- **Sampling Distribution of the Mean:**

Sampling distribution of means of a population data is defined as the **probability distribution of the sample means**, which are obtained by extracting all the possible samples having the same size from the given population.

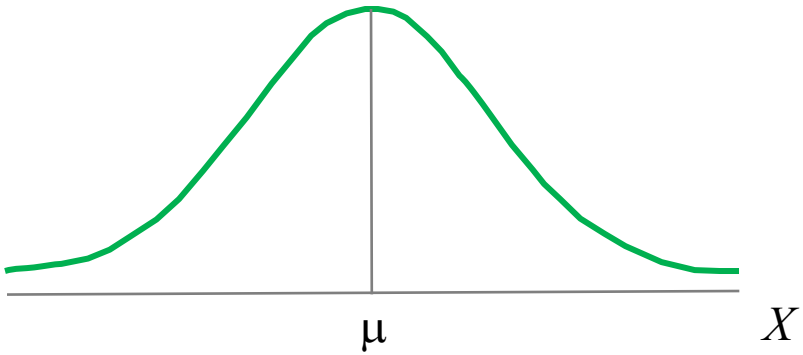
- Given a finite population with mean (μ) and variance (σ^2).
- When sampling from a normally distributed population, the distribution of the sample mean will have the following properties:

SAMPLING DISTRIBUTION PROPERTIES

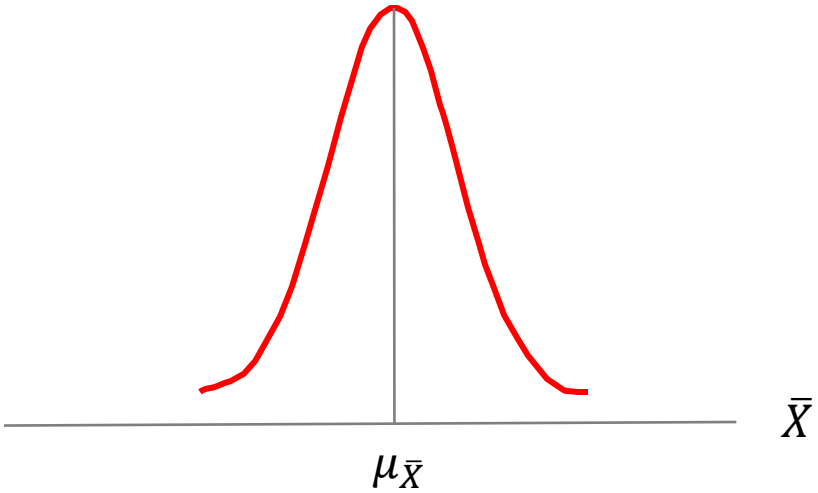
- The distribution of sample mean $\bar{X}$ will be normal
- The mean $\mu_{\bar{X}}$ of the distribution of the values of will be the same as the mean of the population from which the samples were drawn;

$$\mu_{\bar{X}} = \mu$$

Normal Population Distribution



Normal Sampling Distribution (has the same mean)



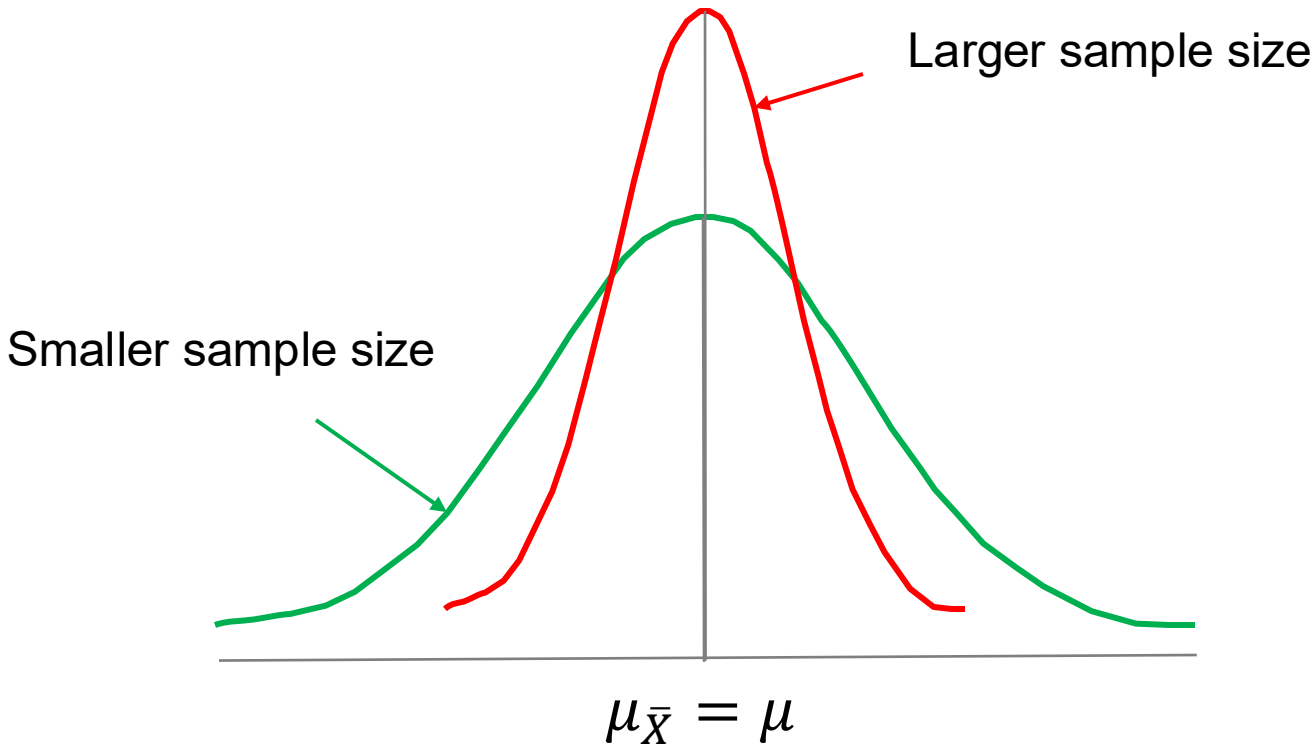
SAMPLING DISTRIBUTION PROPERTIES

- The variance of the distribution of $\bar{X}$, $V(\bar{X})$, $\sigma_{\bar{X}}^2$, will be equal to the variance of the population divided by the sample size;

$$V(\bar{X}) = \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

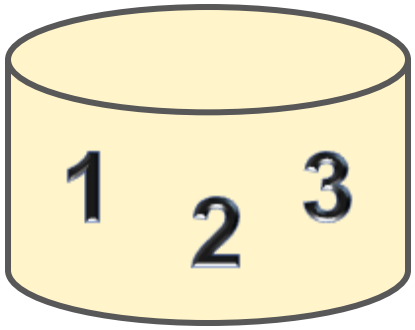
SAMPLING DISTRIBUTION PROPERTIES

- Size of sample size: n



EXAMPLE : SAMPLING DISTRIBUTION

- Population consists of three number : 1, 2 and 3.



MEAN & VARIANCE OF POPULATION

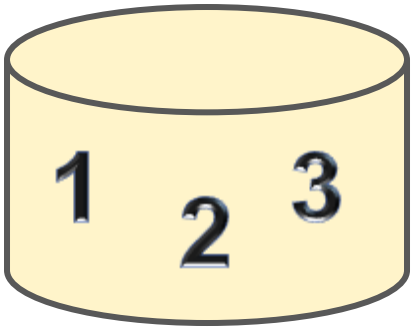
	X	Frequency	Prob.
	1	1	1/3
	2	1	1/3
	3	1	1/3
Mean (μ)	**2**		
Variance (σ^2)	**2/3**		

Population Mean = $\mu = 2$

Population Variance = $\sigma^2 = \frac{(1-2)^2+(2-2)^2+(3-2)^2}{3} = \frac{2}{3}$

EXAMPLE: SAMPLING DISTRIBUTION

- Two numbers are selected at random with replacement and the sequence of numbers in noted.



SAMPLING DISTRIBUTION OF THE SAMPLE MEAN

Total = 3x3 = 9 possibility

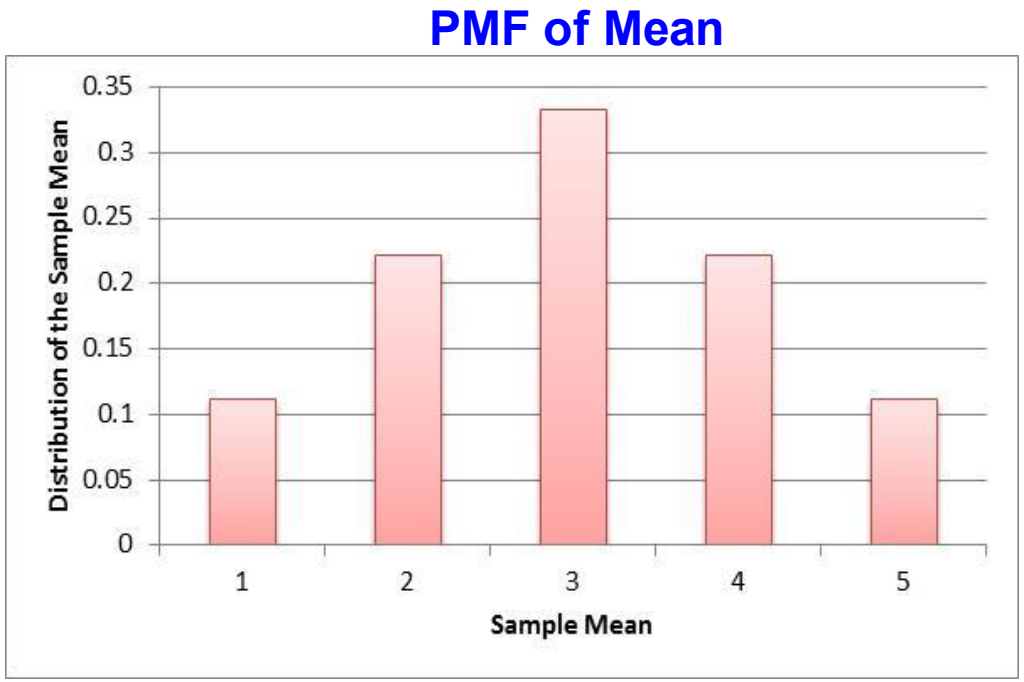
	X_1	X_2	Sample Mean $\bar{X}$
Sample 1	1	1	1
Sample 2	1	2	1.5
Sample 3	1	3	2
Sample 4	2	1	1.5
Sample 5	2	2	2
Sample 6	2	3	2.5
Sample 7	3	1	2
Sample 8	3	2	2.5
Sample 9	3	3	3
Mean $\mu_{\bar{X}}$			2

- **Mean of Sample Mean**

$$\mu_{\bar{X}} = \mu$$

MEAN OF THE SAMPLING DISTRIBUTION OF THE SAMPLE MEAN

Mean $\bar{X}$	Frequency	Prob.
1	1	1/9
1.5	2	2/9
2	3	3/9
2.5	2	2/9
3	1	1/9



$$E(\bar{X}) = \mu_{\bar{X}} = \frac{1 + 1.5 + 1.5 + 2 + 2 + 2 + 2.5 + 2.5 + 3}{9} = 2$$

$$E(\bar{X}) = \mu_{\bar{X}} = \sum \bar{x}_i P(\bar{x}) = 1 \times \frac{1}{9} + 1.5 \times \frac{2}{9} + 2 \times \frac{3}{9} + 2.5 \times \frac{2}{9} + 3 \times \frac{1}{9} = 2$$

VARIANCE OF THE SAMPLING DISTRIBUTION OF THE SAMPLE MEAN

$$\sigma_{\bar{X}}^2 = \frac{(1-2)^2+(1.5-2)^2+(1.5-2)^2+(2-2)^2+(2-2)^2+(2-2)^2+(2.5-2)^2+(2.5-2)^2+(3-2)^2}{9} = \frac{1}{3}$$

$$\sigma_{\bar{X}}^2 = E[(\bar{x} - E(\bar{x}))^2] = E(\bar{x}^2) - [E(\bar{x})]^2$$

$$E(\bar{X}) = 2$$

$$E(\bar{x}^2) = \sum \bar{x}_i{}^2 P(\bar{x}_i)$$

$$= (1^2 \times \frac{1}{9}) + (1.5^2 \times \frac{2}{9}) + (2^2 \times \frac{3}{9}) + (2.5^2 \times \frac{2}{9}) + (3^2 \times \frac{1}{9}) = \frac{13}{3}$$

$$\sigma_{\bar{X}}^2 = \frac{13}{3} - 4 = \frac{1}{3}$$

$$\sigma^2 = \frac{2}{3}, \ \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n=2} = \frac{2}{3\times 2} = \frac{1}{3}$$

EXAMPLE : SAMPLING DISTRIBUTION OF MEANS

A population consists of five numbers 2, 3, 6, 8, and 11. Consider all possible samples of size 2 that can be drawn with and without replacement from this population.

a) The mean of the population.

b) The standard deviation of the population.

c) The mean of the sampling distribution of means.

d) Standard deviation of the sampling distribution of means (the standard error of means)

SOLUTION

a) Mean of the population μ = (2+3+6+8+11)/5 = 6

b) Standard deviation of population

$$\sigma^2 = \frac{(2-6)^2+(3-6)^2+(6-6)^2+(8-6)^2+(11-6)^2}{5} = 10.8$$

$$\sigma = 3.29$$

Sampling size 2, With replacement :

c) There are 25 samples of size 2 **Corresponding sample means**

(2,2)	(2,3)	(2,6)	(2,8)	(2,11)	2.0	2.5	4.0	5.0	6.5
(3,2)	(3,3)	(3,6)	(3,8)	(3,11)	2.5	3.0	4.5	5.5	7.0
(6,2)	(6,3)	(6,6)	(6,8)	(6,11)	4.0	4.5	6.0	7.0	8.5
(8,2)	(8,3)	(8,6)	(8,8)	(8,11)	5.0	5.5	7.0	8.0	9.5
(11,2)	(11,3)	(11,6)	(11,8)	(11,11)	6.5	7.0	8.5	9.5	11.0

SOLUTION(CONT.)

the mean of sampling distribution of means is

$$\mu_{\bar{X}} = \frac{\text{sum of all sample means}}{25} = \frac{150}{25} = 6.0$$

d) Standard deviation of the sampling distribution of mean

$$\sigma^2{}_{\bar{X}} = \frac{(2-6)^2+(2.5-6)^2+(2.5-6)^2+\cdots+(11-6)^2}{25} = \frac{135}{25}$$

$$= \mathbf{5.40} = \frac{\sigma^2}{n} = \frac{10.8}{2}$$

$$\therefore \sigma_{\bar{X}} = 2.32$$

SOLUTION(CONT.)

- **X: Select 2 sample without replacement:**

c) There are 10 samples of size 2: 5C2

(2,3) (2,6) (2,8) (2,11)
(3,6) (3,8) (3,11)
(6,8) (6,11)
(8,11)

The corresponding sample means are :

2.5, 4.0, 5.0, 6.5, 4.5, 5.5, 7.0, 7.0, 8.5, 9.5

SOLUTION(CONT.)

The mean of sampling distribution of means is

$$\mu_{\bar{X}} = \frac{2.5+4.0+5+6.5+4.5+5.5+7.0+7.0+8.5+9.5}{10} = 6.0 = \mu$$

d) The variance of sampling distribution of mean is,

$$\sigma^2{}_{\bar{X}} = \frac{(2.5-6)^2+(4-6)^2+\cdots+(9.5-6)^2}{10} = 4.05$$

$$\therefore \sigma_{\bar{X}} = 2.01$$

Z-VALUE FOR SAMPLING DISTRIBUTION OF THE MEAN

- Z-value for the sampling distribution of $\bar{X}$:

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where $\bar{X}$ = sample mean

μ = population mean

σ = population standard deviation

n = sample size

CENTRAL LIMIT THEOREM

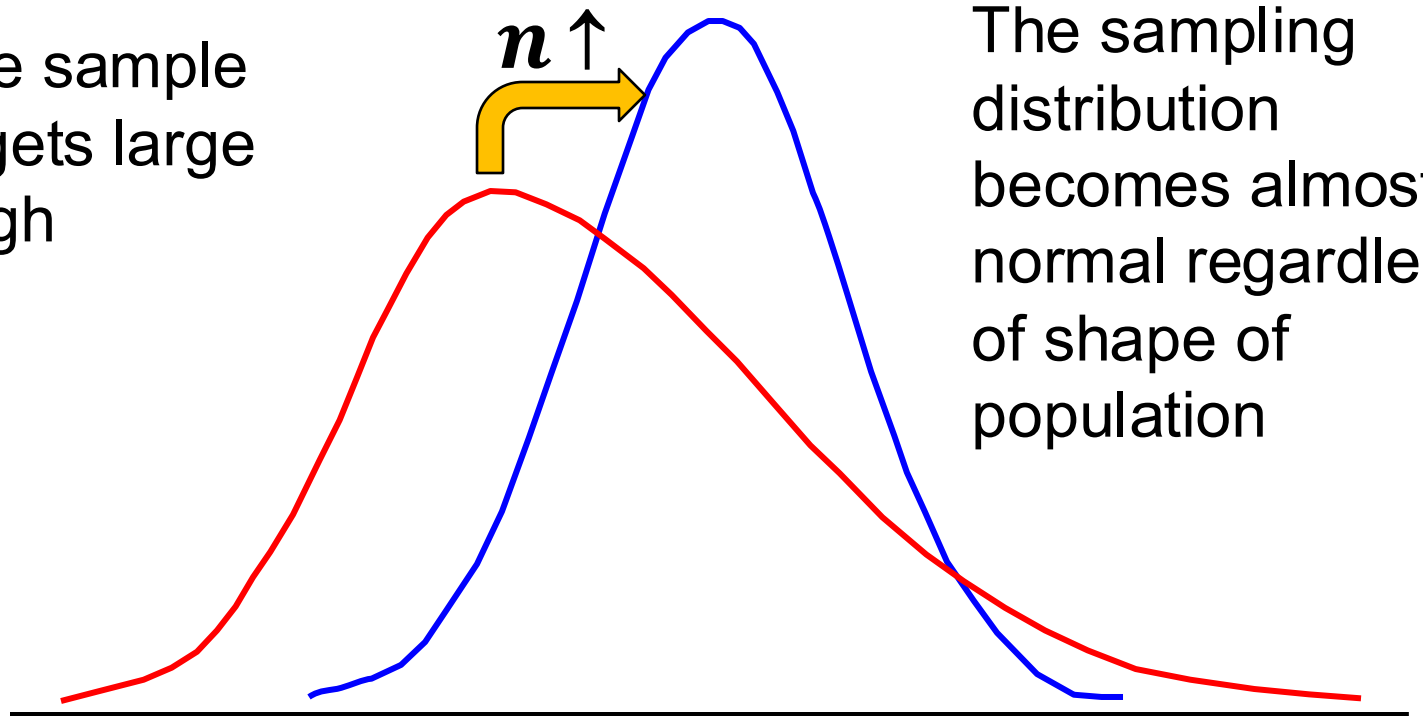
If we select a large number of random samples, e.g., from any population distribution and determine the mean of each sample,

the distribution of these sample means can be described by the normal distribution with a mean of μ and variance σ^2/n.

- **The sampling distribution of sample means approaches to normal distribution.**

CENTRAL LIMIT THEOREM

As the sample size gets large enough



The sampling distribution becomes almost normal regardless of shape of population

For many distribution, $n \geq 30$ will give a sampling distribution becomes nearly normal distribution

EXAMPLE

- Suppose a population has mean μ = 8 and standard deviation σ =3. A random sample of size n = 36 is selected.
- What is the probability that the sample mean is between 7.8 and 8.2?

SOLUTION

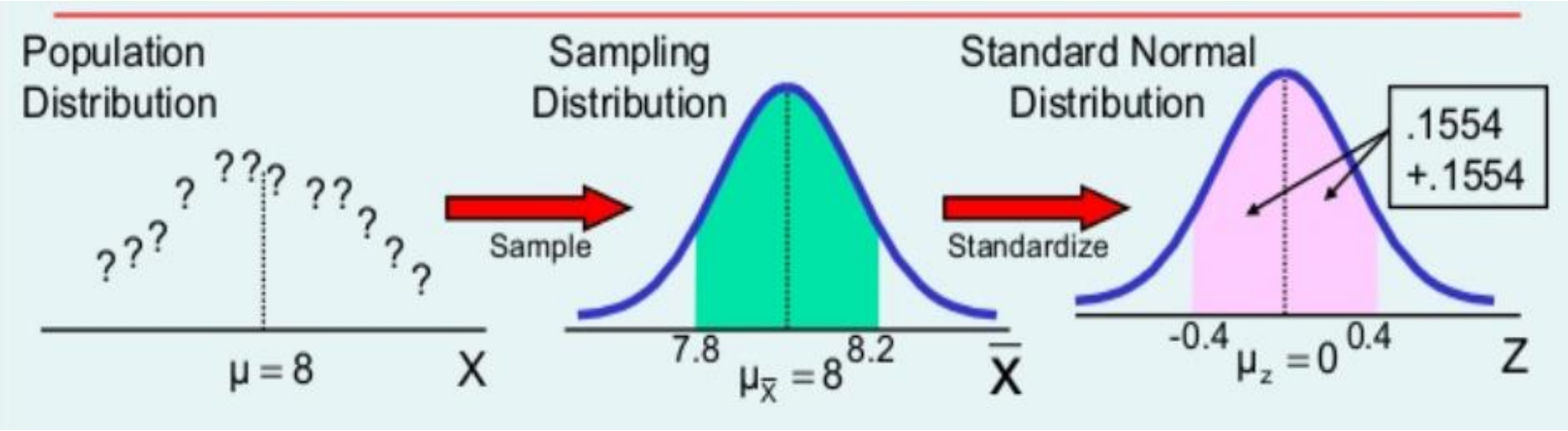
- Even the population is not normally distribution, the central limit theorem can be use (n > 30) so the sampling distribution of is approximately normal with mean $\mu_{\bar{X}} = 8$ and standard deviation $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$

- $P(7.8 < \bar{X} < 8.2) = P(Z_1 < Z < Z_2)$

- $P(7.8 < \bar{X} < 8.2) = P\left(\frac{7.8-8}{3/\sqrt{36}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{78.2-8}{3/\sqrt{36}}\right)$

  $P = (-0.4 < Z < 0.4) = 0.3108$

SOLUTION (CONT..)



TYPE OF SAMPLING DISTRIBUTION

- **Sampling distribution of proportion**
- **Population proportions**
  - p = proportion of the **population** that contains some desired characteristic
  - Sample proportion ($\hat{p}$) provides an estimation of p
  - $\hat{p} = \frac{X}{n} = \frac{\text{number of success sample}}{\text{sample size}}$
  - $0 \leq \hat{p} \leq 1$
  - $\hat{p}$ is approximately normal distributed when n is large

SAMPLING DISTRIBUTION OF PROPORTION

- If the sample size is much smaller than the size of a population with proportion p of successes, then the estimate proportion $\hat{p}$ can be approximate by normal distribution with mean $\mu_{\hat{p}}$ and standard deviation $\sigma_{\hat{p}}$

$$\mu_{\hat{p}} = p$$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Z-VALUE FOR PROPORTION

- Standardize $\hat{p}$ to a Z-value with the form

$$Z = \frac{\hat{p} - p}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

EXAMPLE

- If the true proportion of voters who support Team-A is $p = 0.4$
- What is the probability that a sample of size 200 yields a sample proportion between 0.40 and 0.45?

SOLUTION

If $p = 0.4$ and n = 200, What is $P(0.40 \leq \hat{p} \leq 0.45)$?

Find $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.4(1-0.4)}{200}} = 0.03464$

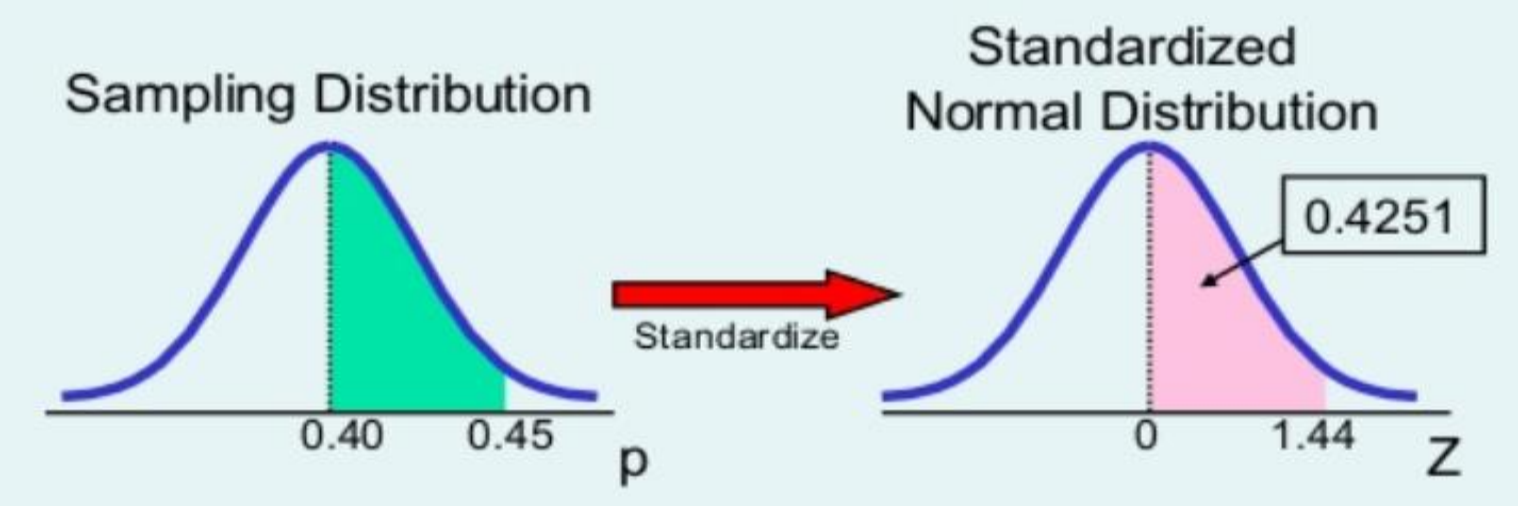
$$P(0.40 \leq \hat{p} \leq 0.45) = P\left(Z_1 \leq Z_{\hat{p}} \leq Z_2\right)$$

$$Z_1 = \frac{0.40-0.40}{0.03464} = 0\ , Z_2 = \frac{0.45-0.40}{0.03464} = 1.44$$

$$P(0.40 \leq \hat{p} \leq 0.45) = P(0 \leq Z \leq 1.44) = 0.4251$$

SOLUTION(CONT.)

$$P(0 \leq Z \leq 1.44) = 0.4251$$



STANDARD ERROR OF THE SAMPLING DISTRIBUTION

The sampling distribution has a standard deviation.

- Even the mean of the sampling distribution is the same as the population mean $E(\bar{X}) = \mu$, but the standard deviation of sampling distribution may differ from the population standard deviation.

STANDARD ERROR OF THE SAMPLING DISTRIBUTION

The variation of sample mean around the population mean is the **sampling error** and is measured using a statistic known as the standard error of the mean.

Standard error of the mean = an estimate of the amount that a sample mean is likely to differ from the population mean.

- **Standard Error**
  - Standard error of the sample mean
  - Standard error of the sample proportion

STANDARD ERROR OF THE MEAN

The standard error of a sampling distribution is equal standard deviation of sampling distribution

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$\sigma_{\bar{X}}$ = Standard deviation of sample mean.

σ = Standard deviation of population

n = Sample size

STANDARD ERROR OF THE PROPORTION

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

$\sigma_{\hat{p}}$ = Standard error for sample Proportion

p = Population Proportion

n = number of samples from population

EXAMPLE : SAMPLING DISTRIBUTION OF PROPORTIONS

Find the probability that in 120 tosses of a fair coin,

a) Between 40% and 60% of tosses will be heads

b) 5/8 or more will be heads

SOLUTION

a) Between 40% and 60% will be heads and

$p = 0.5$, n = 120, $\hat{p}_1 = 40\%$, $\hat{p}_2 = 60\%$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.5(1-0.5)}{120}} = 0.0456$$

$P(0.40 \leq \hat{p} \leq 0.6)$

$= P\left(\frac{0.40-0.50}{0.0456} \leq Z \leq \frac{0.60-0.50}{0.0456}\right)$

$= P(-2.19 \leq Z \leq 2.19)$

$= 0.98574 - 0.01426 = 0.97148$

b) 5/8 or more will be heads

$$P(\hat{p} \geq 0.625) = P\left(Z \geq \frac{0.625 - 0.5}{0.0456}\right) = P(Z \geq 2.74)$$

$$= 1 - P(Z \leq 2.74) = 1 - 0.99693 = 0.00307$$

ESTIMATION

- There are two types of estimates of population parameter statistics :
  - **Point Estimation** is a single number, calculated from available sample data, used to estimate the value of an unknown population parameter.
  - **Interval Estimate (Confidence Interval) :** is an interval that provides **an upper and lower bound for a specific** unknown population parameter.

Point estimate

- Sample mean
- Sample proportion

Interval estimate

- Confidence interval for mean
- Confidence interval for proportion

POINT ESTIMATION

- A point estimate of a population parameter is **a single value**

For example,

- The sample mean $\overline{X}$ is point estimate of the population mean μ.

- The sample proportion $\hat{p}$ is a point estimate of the population proportion p.

POINT ESTIMATION

Ex, We wish to estimate, the average population household expenditure.

A random sample of independent household expenditure, $X_1, \ldots, X_n$ can be taken.

Consider

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

POINT ESTIMATION

Consider

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

If $n = 10$ households and their expenditures are: $(X_{\mathrm{n}}$: $1874, 1642, 1603, 1931, 2103, 2068, 1948, 1798, 2364, 1918)$, then an estimate of using $\bar{X}$ would be

$$\frac{1874 + 1642 + \cdots + 1918}{10} = 1924.9$$

This type of estimate is called a **point estimate**.

INTERVAL ESTIMATION

The difference between a point estimate and a true parameter is called **sampling error**.

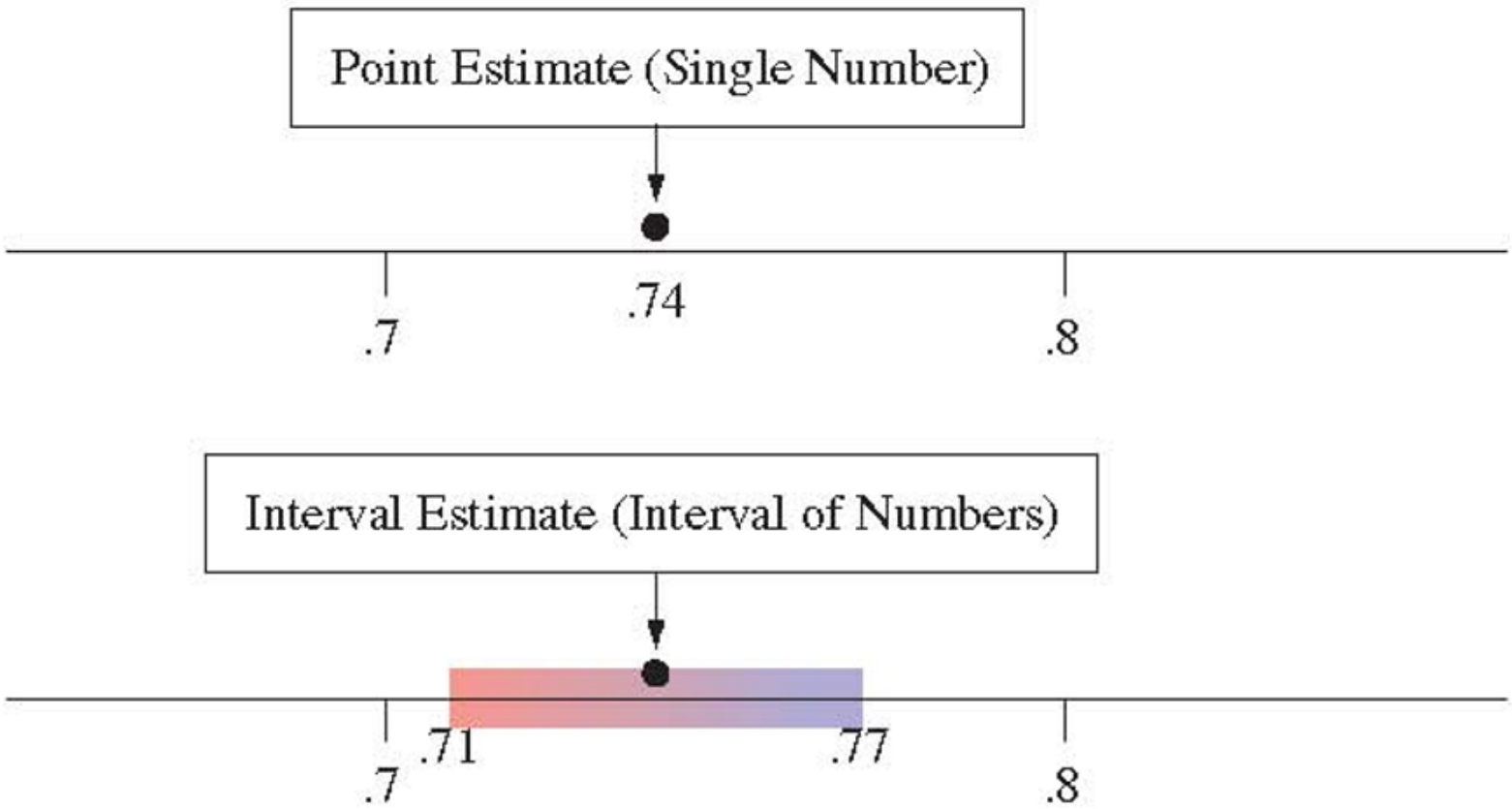
Sampling error arises because we use a sample (only a part of the population) to infer about the entire population.

When we use $\bar{X}$ to estimate μ, we must account for the sampling error. Instead of saying 1924.9 or 1995 with no reference to our level of belief of our estimate, *we say*

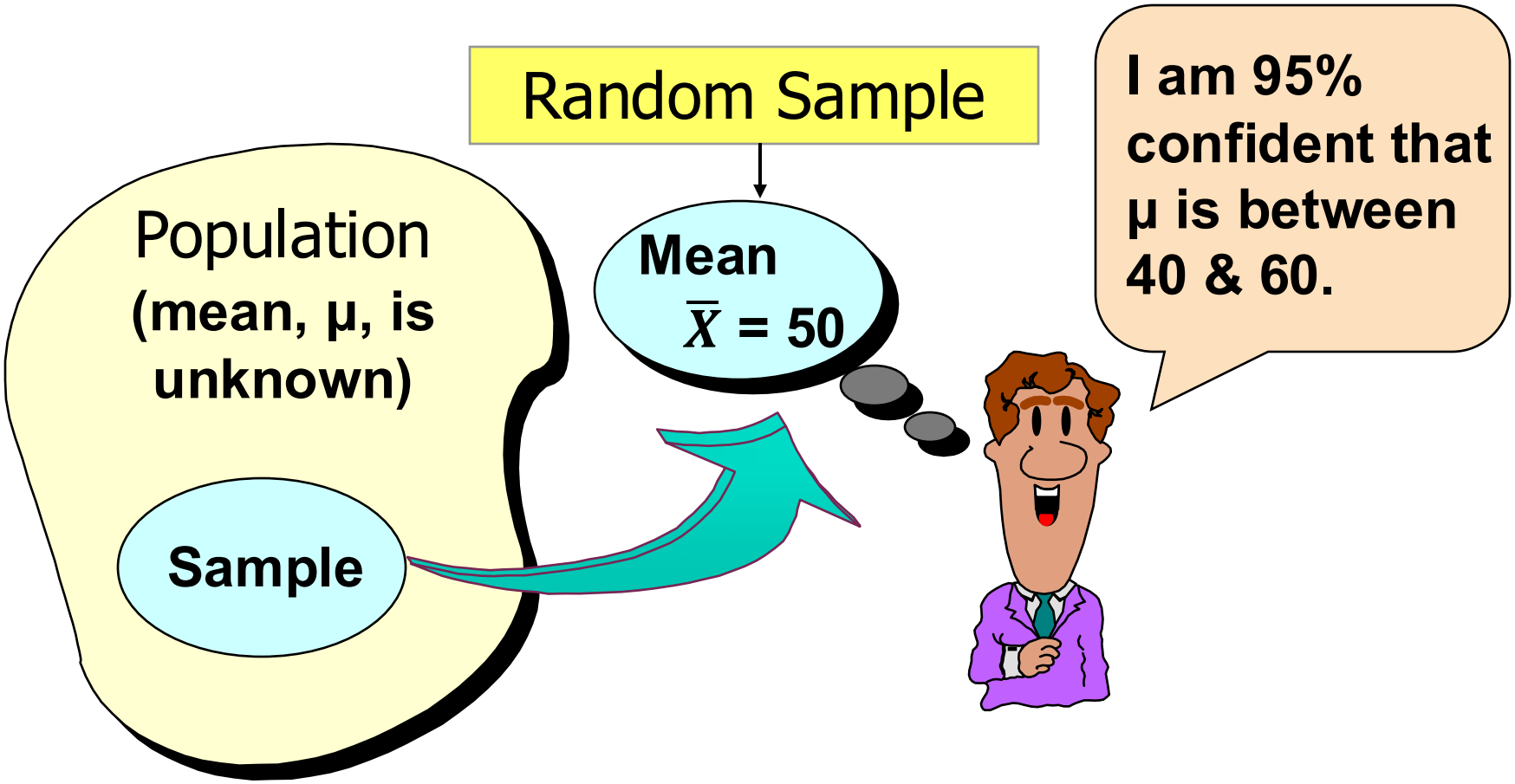
μ is within (a, b) with k% confidence.

This is called a **confidence interval estimate**

INTERVAL ESTIMATE

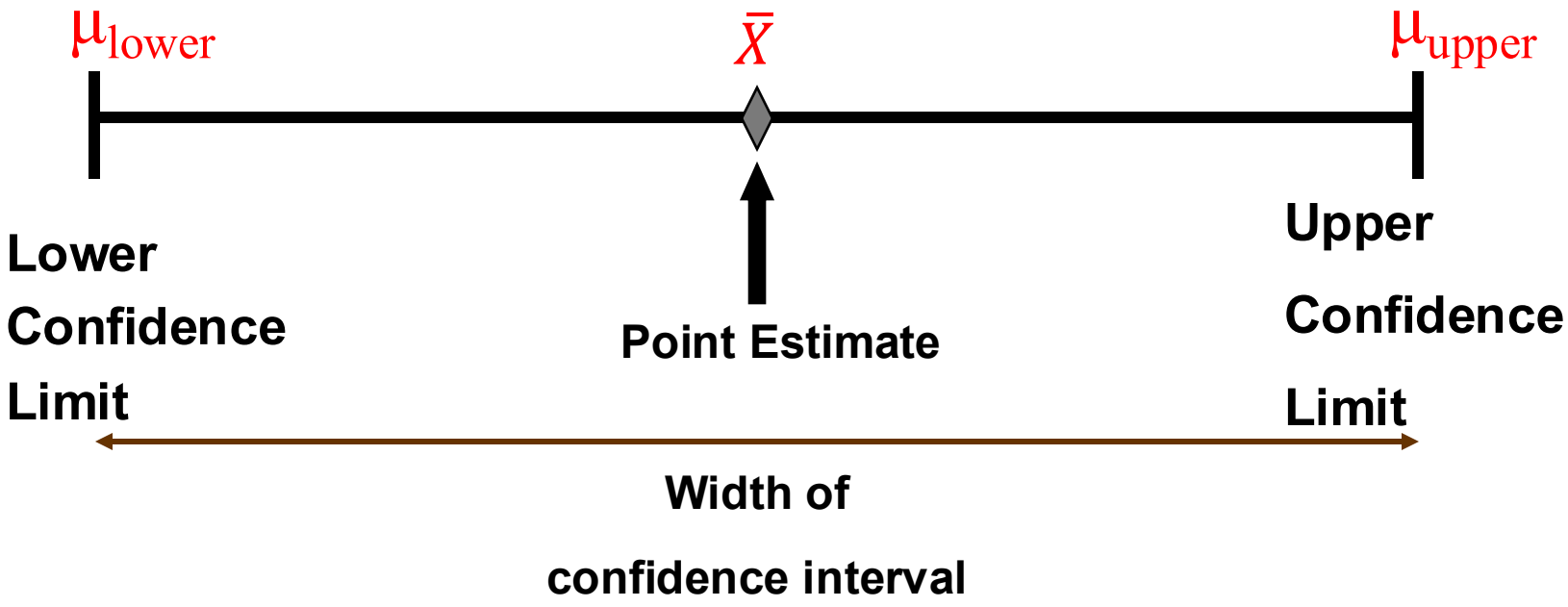


ESTIMATION PROCESS



POINT AND INTERVAL ESTIMATION

Interval Estimation: a range of numbers within which the parameter (estimator) is believed to fall (lower bound, upper bound)

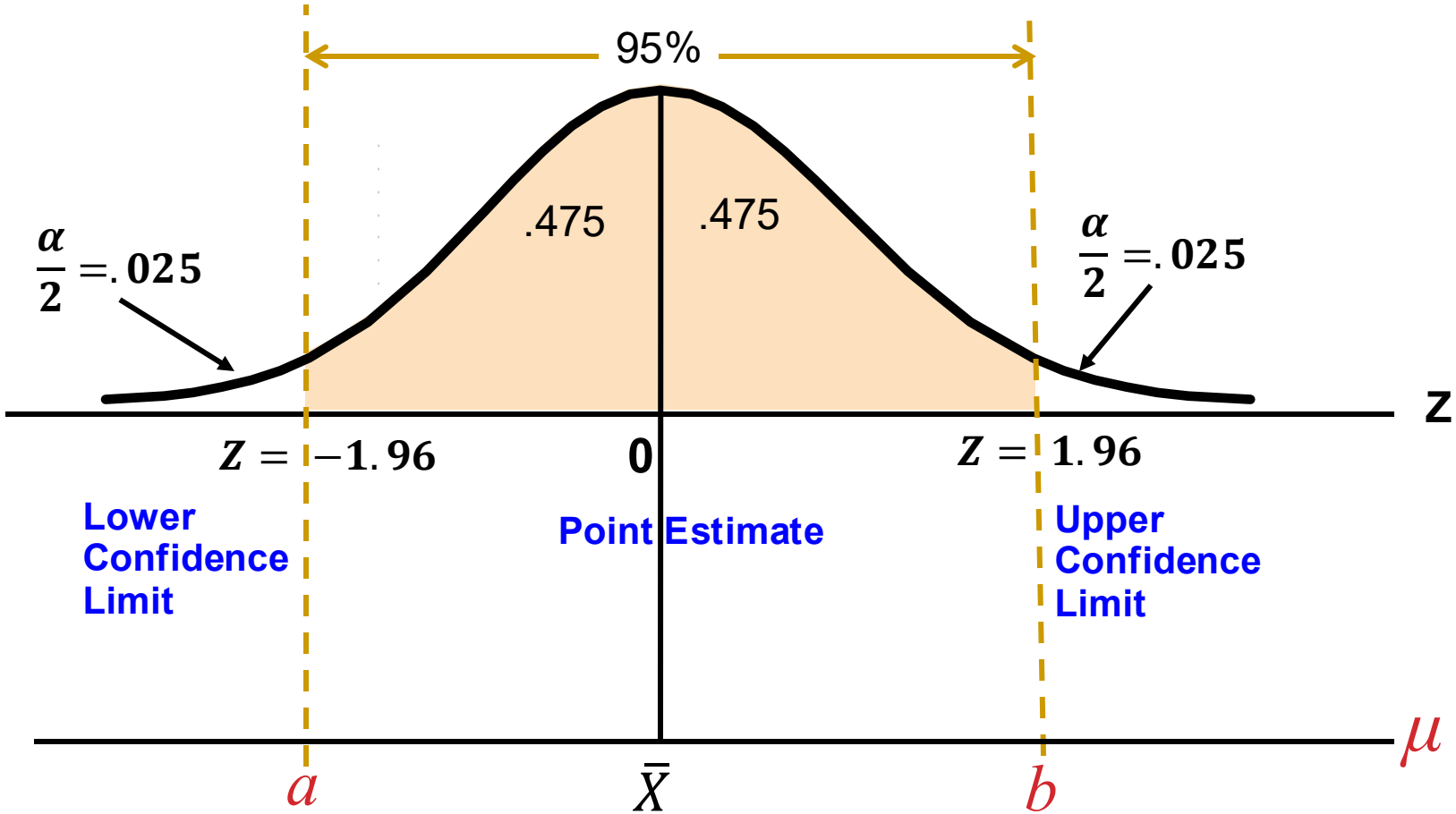


CONFIDENCE INTERVAL

- It involves
  - **Confidence coefficient**: the probability that the interval contains the parameter, e.g., 0.90, 0.95, 0.99. or …
  - **Confidence level ($1 - \alpha$)**: 90%, 95%
  - **Error probability (α)**: the probability that an interval does not contain the parameter.

EXAMPLE : 95% CONFIDENCE LEVEL

- Consider a 95% confidence interval:



DEVELOPMENT OF THE CONFIDENCE INTERVAL AND ITS BASIC PROPERTIES

$\bar{X}$ is normally distributed with mean μ and variance σ^2/n.

The standard normal distribution of Z from $\bar{X}$ is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

A confidence interval estimate for μ is

$$l < \mu < u$$

The probability is

$$P[l < \mu < u] = 1 - \alpha$$

$$P\left[-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right] = 1 - \alpha$$

$$P\left[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

CONFIDENCE INTERVAL

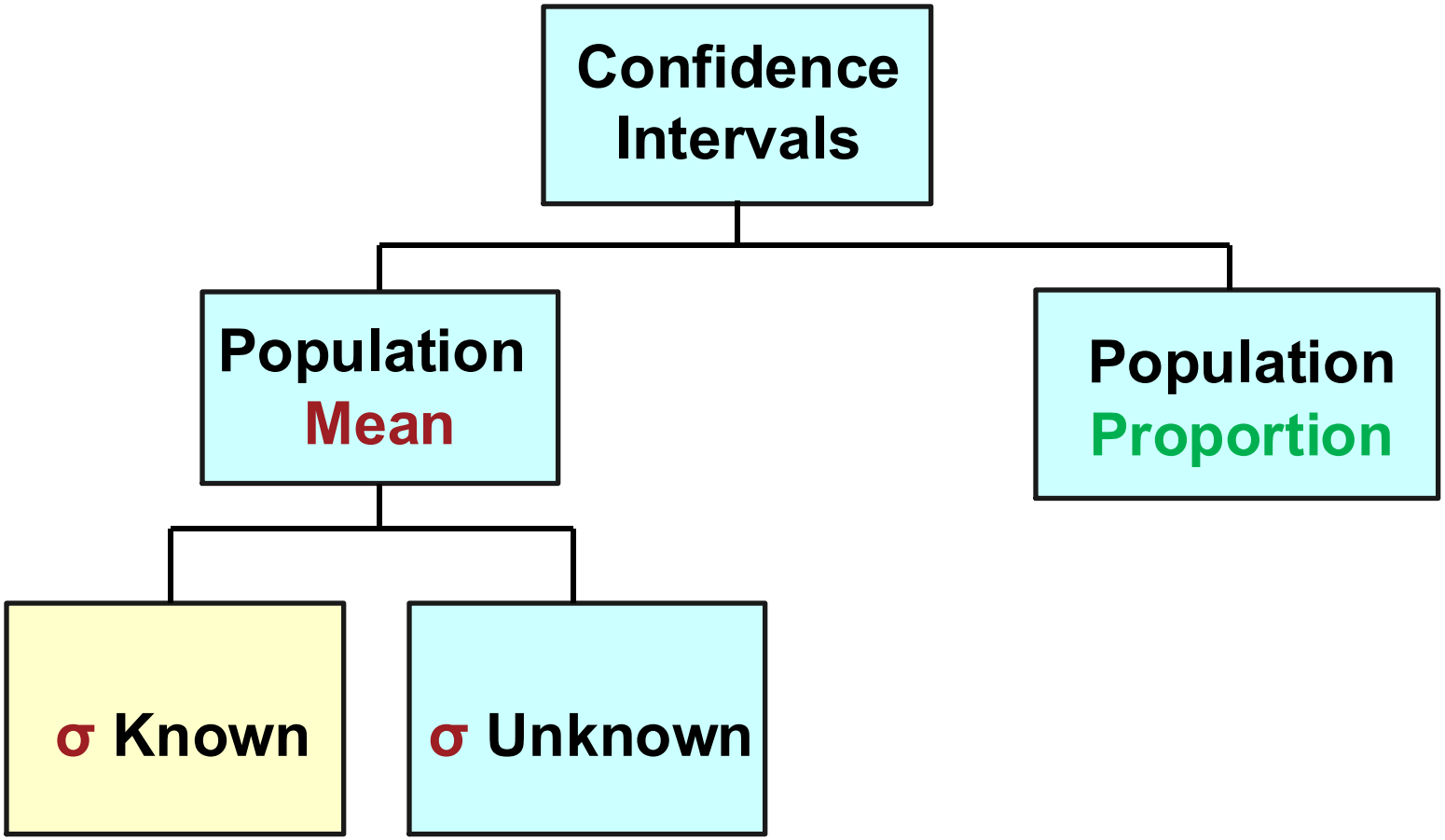
confidence Interval

"A range of values around a point estimate that makes it possible to state the probability that an interval contains the population parameter between its lower and upper bounds."

The general formula for all confidence intervals is:

Point Estimate ± (Critical Value)(Standard Error)

CONFIDENCE INTERVALS



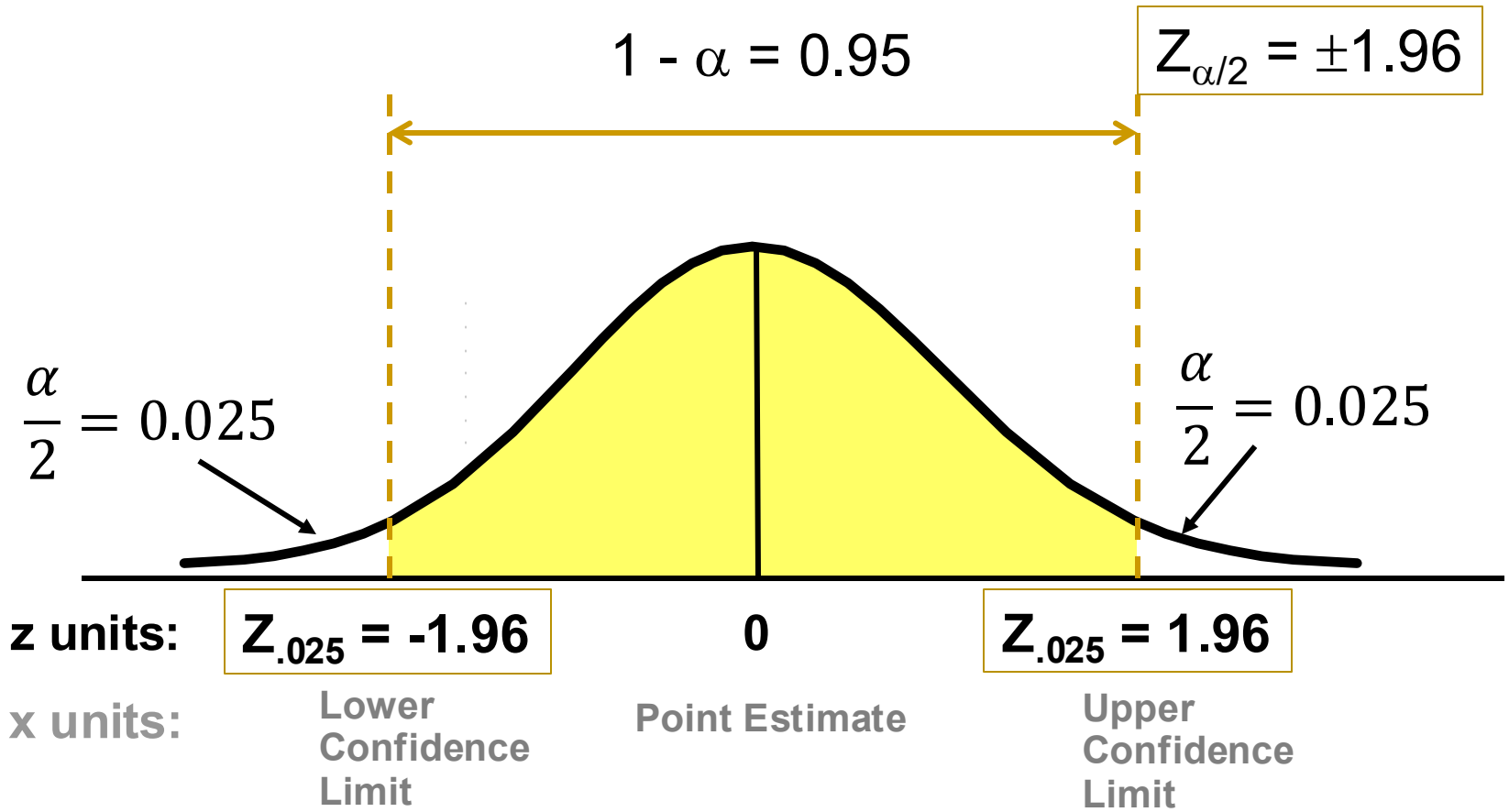
CONFIDENCE INTERVAL FOR μ (σ KNOWN)

- Assumptions
  - Population standard deviation σ is known
  - Population is normally distributed
  - If population is not normal, use large sample
- Confidence interval estimate for μ

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

FINDING THE CRITICAL VALUE

- Consider a 95% confidence interval:



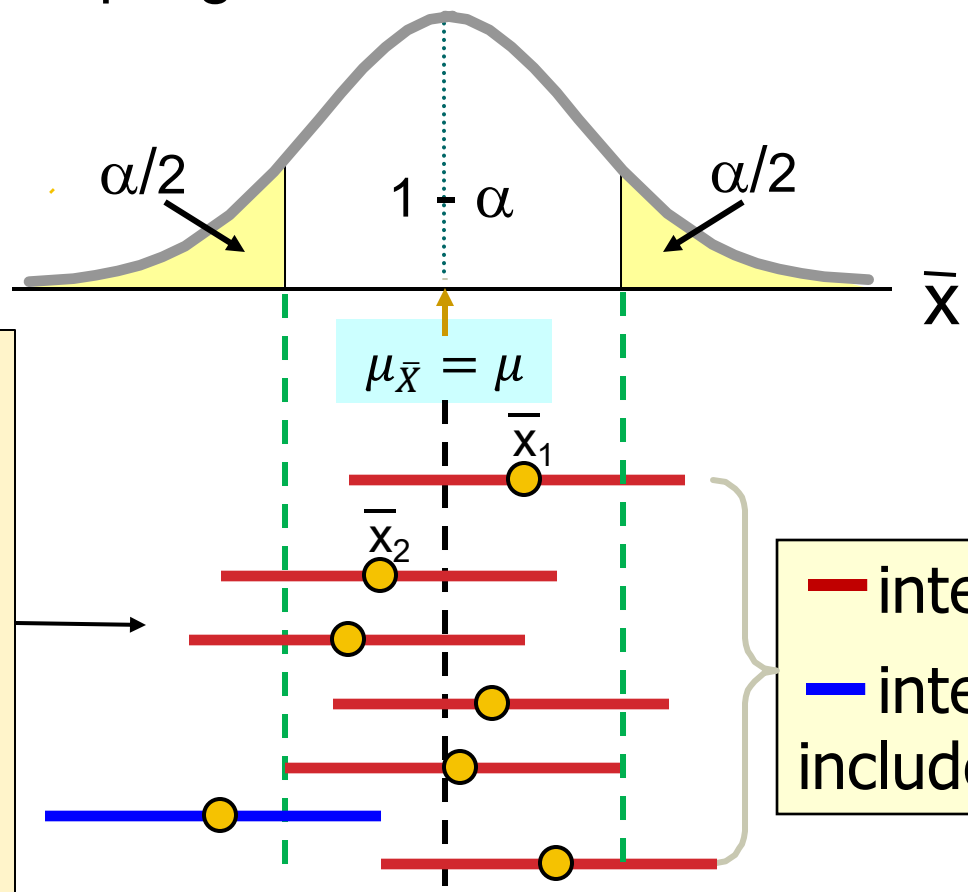
COMMON LEVELS OF CONFIDENCE

- Commonly used confidence levels are 90%, 95%, and 99%

Confidence Level	Confidence Coefficient, $1 - \alpha$	z value, $Z_{\alpha/2}$
80%	0.80	1.28
90%	**0.90**	**1.645**
95%	**0.95**	**1.96**
98%	0.98	2.33
99%	**0.99**	**2.57**
99.8%	0.998	3.08
99.9%	0.999	3.27

INTERVAL AND LEVEL OF CONFIDENCE

Sampling Distribution of the Mean



Intervals Extend from

$$\bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

to

$$\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

interval include μ

interval dose not include μ

Confidence Intervals

MARGIN OF ERROR

- **Margin of Error (e):** the amount added and subtracted to the point estimate to form the confidence interval

Example: Margin of error for estimating μ, σ known:

$$\bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \qquad e = Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

- Interval Estimate of μ

$$\boxed{\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}$$

where:

$\bar{X}$	is the sample mean
1 - α	is the confidence coefficient
$z_{\alpha/2}$	is the *z* value providing an area of $\alpha/2$ in the upper tail of the standard normal probability distribution
σ	is the population standard deviation
n	is the sample size

FACTORS AFFECTING MARGIN OF ERROR

$$e = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Intervals Extend from
$\bar{X} - Z\sigma_{\bar{X}}$ to $\bar{X} + Z\sigma_{\bar{X}}$

- Data variation, σ : e ⇩ as σ ⇩
- Sample size, n : e ⇩ as n ⇧
- Level of confidence, 1 - α : e ⇩ if 1 - α ⇩

EXAMPLE

A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is .35 ohms.

- Determine a 95% confidence interval for the true mean resistance of the population.

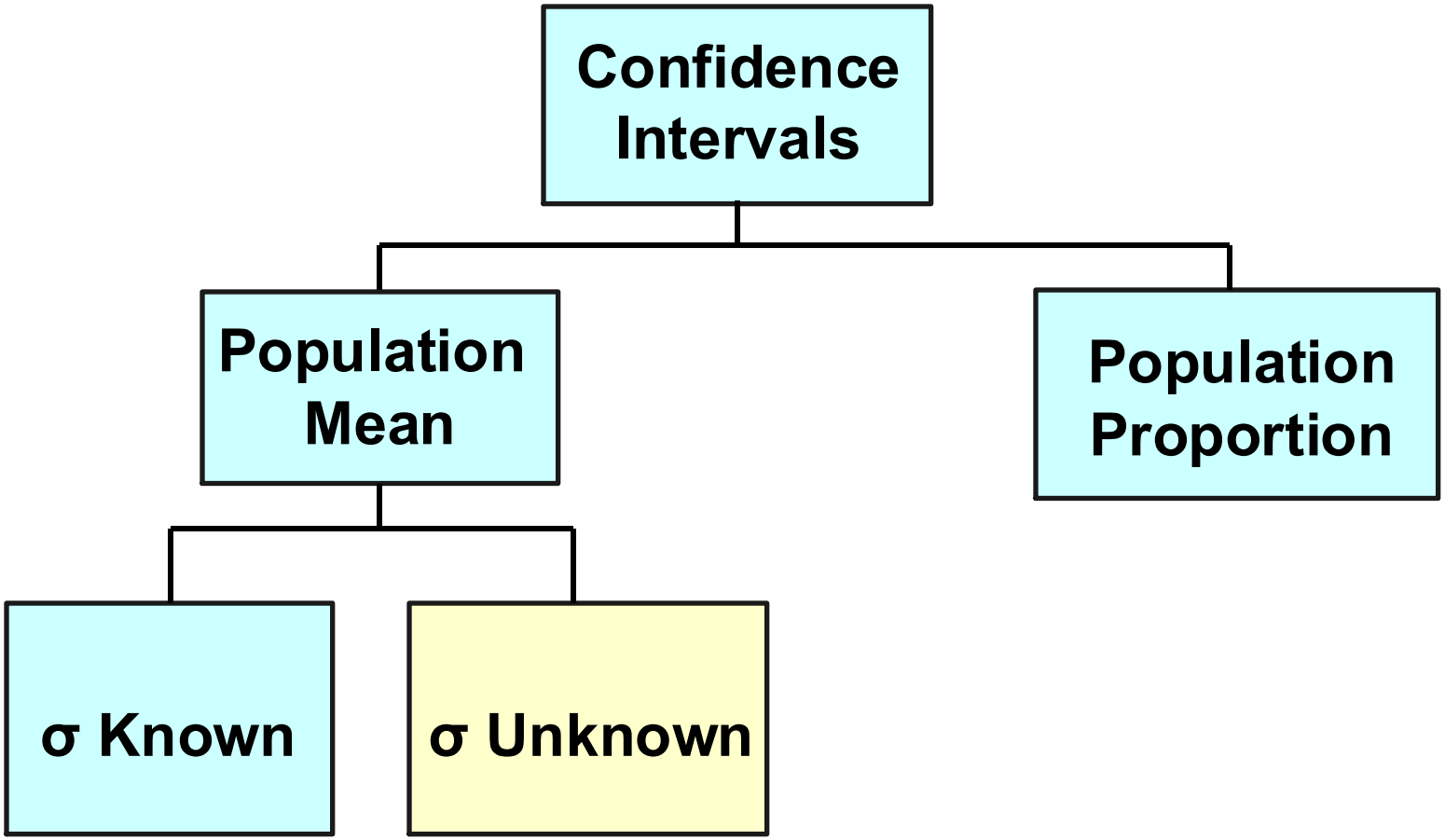
Solution

$$\bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 2.20 \pm 1.96\left(\frac{0.35}{\sqrt{11}}\right)$$

$$= 2.20 \pm 0.2068$$

$$= (1.9932, 2.4068)$$

CONFIDENCE INTERVALS



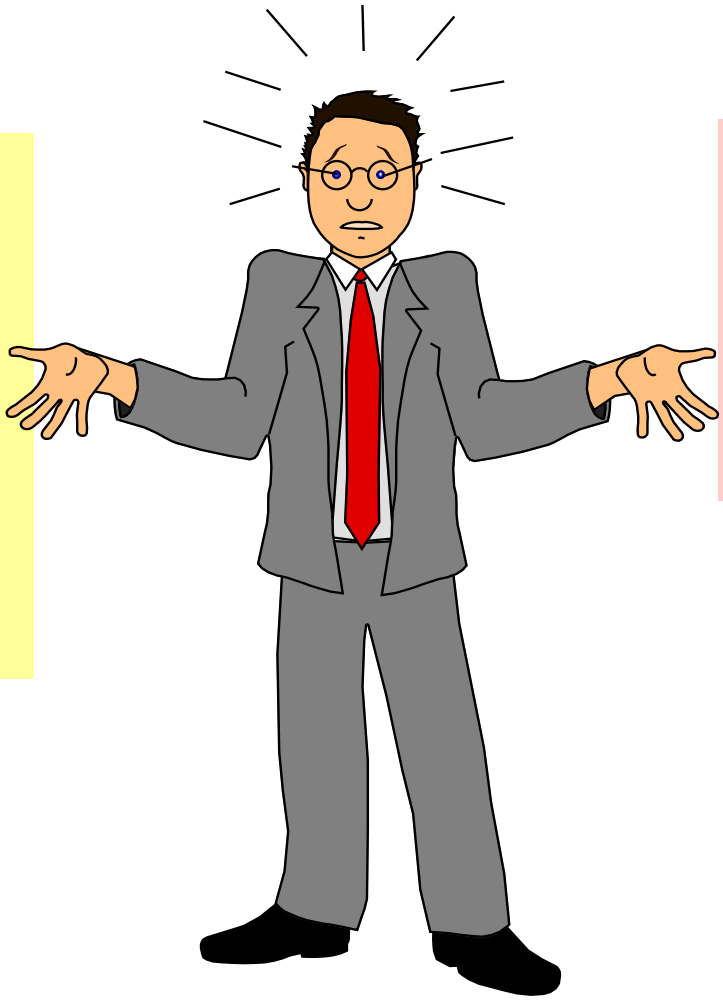
CONFIDENCE INTERVAL FOR μ (σ UNKNOWN)

- If the population standard deviation σ is unknown, we can substitute the sample standard deviation, s
- This introduces extra uncertainty, since s is variable from sample to sample
- So, we use the t-distribution instead of the normal distribution

DETERMINING SAMPLE SIZE (COST)

Too Big:

- Requires too many resources



Too small:

- Won't do the job

CONFIDENCE INTERVAL FOR MEAN μ (σ UNKNOWN)

- Assumptions
  - Population standard deviation is unknown
  - Population is normally distributed
  - If population is not normal, use large sample n
- Use Student's *t*-Distribution
- Confidence Interval Estimate

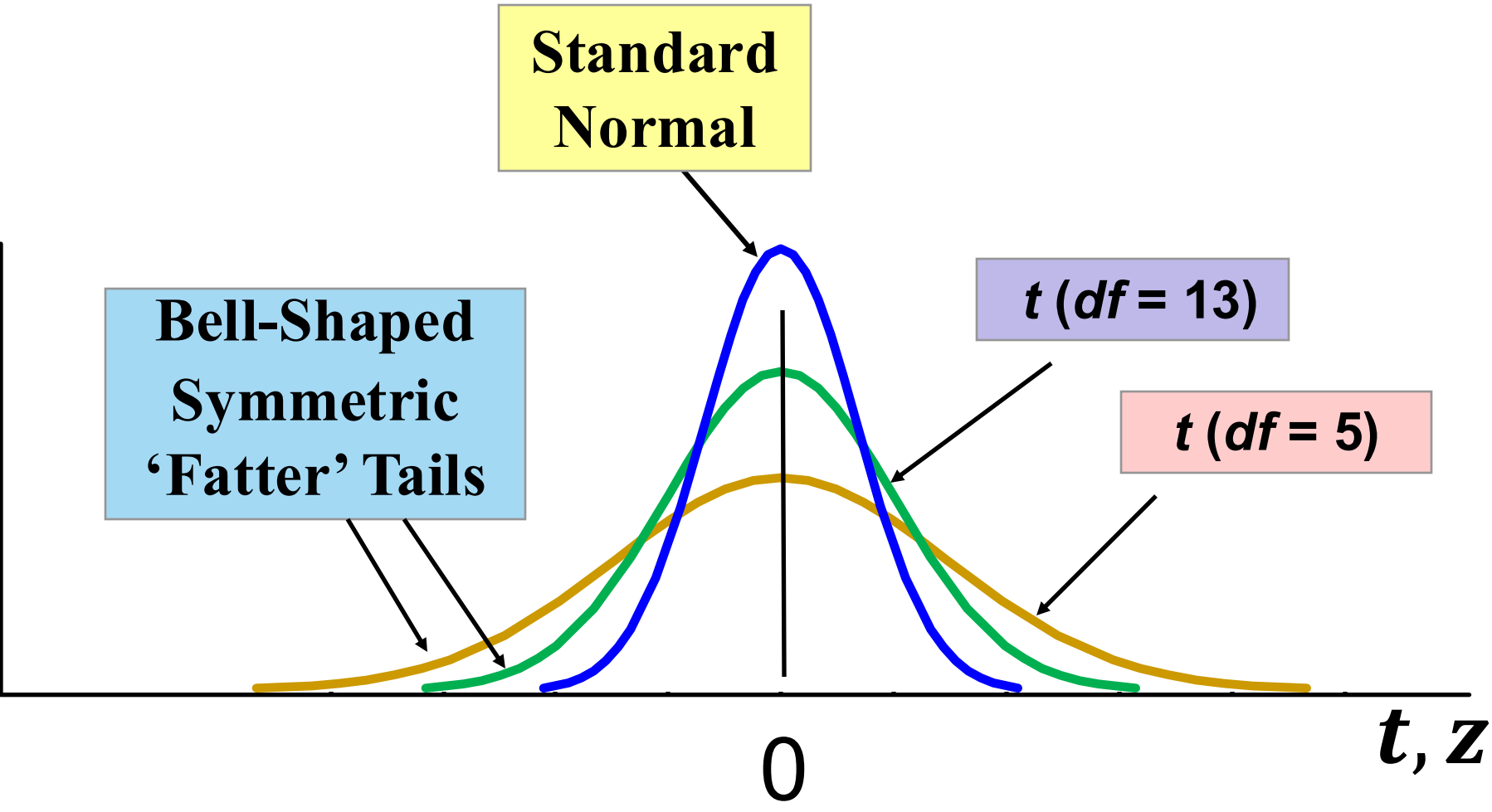
$$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

APPROXIMATION FOR LARGE SAMPLES

- Since t-distribution approaches z-distribution as the sample size n increases, an approximation is sometimes used when $n \geq 30$

Technically correct	Approximation for large n
$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$

STUDENT'S T-DISTRIBUTION



STUDENT'S T-DISTRIBUTION

$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

ν :Degree of freedom

$$\Gamma(z) = \int_0^\infty e^{-t}t^{z-1}dt$$: Gamma Function

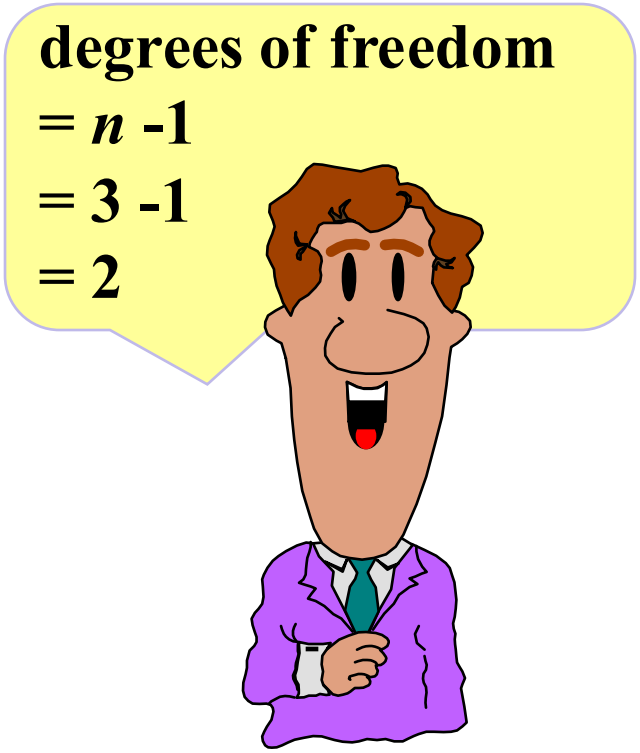
DEGREES OF FREEDOM (*DF*)

- Number of observations that are free to vary after sample mean has been calculated
- Example:
  - Mean of 3 numbers is 2

$$X_1 = 1 \text{ (or any number)}$$

$$X_2 = 2 \text{ (or any number)}$$

$$X_3 = 3 \text{ (cannot vary)}$$



STUDENT'S t TABLE

Let: $n = 3$

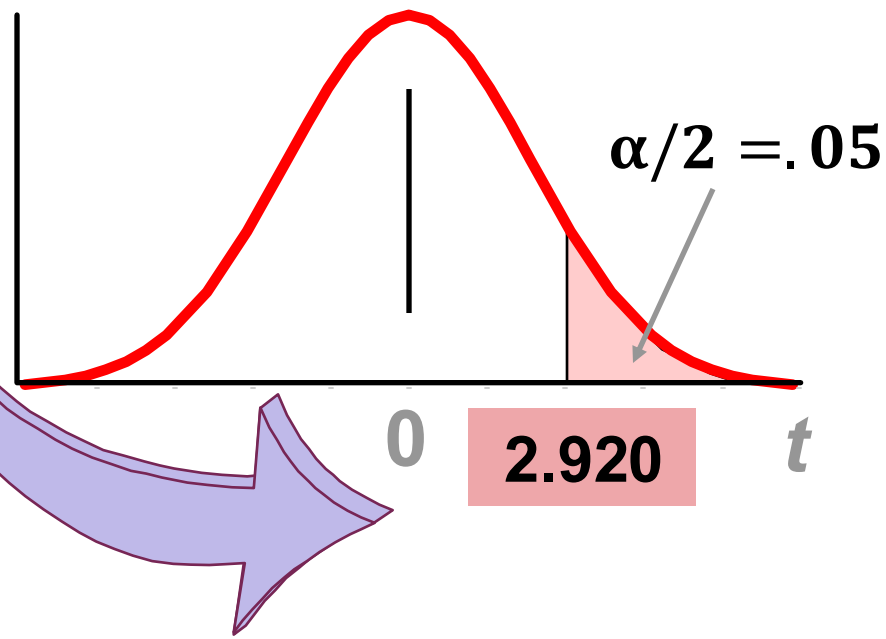
df = n - 1 = 2

$\alpha = .10$

$\alpha/2 = .05$

	Upper Tail Area		
df	0.255	0.10	0.05
1	1.000	3.078	6.314
2	0.817	1.886	2.920
3	0.765	1.638	2.353

t Values



t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

t DISTRIBUTION VALUES

With comparison to the z value

Confidence Level	t (10 d.f.)	t (20 d.f.)	t (30 d.f.)	z
0.80	1.372	1.325	1.310	1.28
0.90	1.812	1.725	1.697	1.64
0.95	2.228	2.086	2.042	1.96
0.99	3.169	2.845	2.750	2.57

Note: t → z as n increases

EXAMPLE

A random sample of $n = 25$ has $\bar{X} = 50$ and $S = 8$. Set up a 95% confidence interval estimate for μ

$$\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$$

$$50 - 2.0639\frac{8}{\sqrt{25}} \leq \mu \leq 50 + 2.0639\frac{8}{\sqrt{25}}$$

$$46.69 \leq \mu \leq 53.30$$

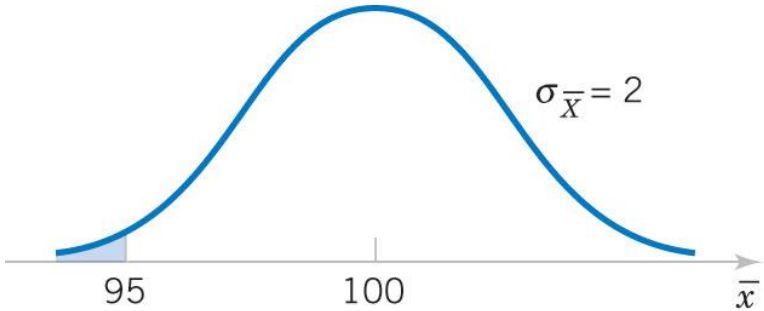
EXAMPLE : RESISTORS

- An electronics company manufactures resistors having a mean resistance of 100 ohms and standard deviation of 10 ohms. The distribution of resistance is normal.
- What is the probability that a random sample of $n = 25$ resistors will have an average resistance less than 95 ohms.

Let X : resistance value

$\mu = 100$, $\sigma = 10$

$P[\bar{X} \leq 95] = ??$



$$s = \sigma_{\bar{X}} = \frac{10}{\sqrt{25}} = 2$$

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{95 - 100}{2} = -2.5$$

$$P[\bar{X} \leq 95] = P[Z \leq -2.5] = 0.0062$$

EXAMPLE

- A lightbulb manufacturer claims that the lifespan of its lightbulb has a mean of 54 months and a standard deviation of 6 months.
- Your consumer advocacy group tests 50 of them. Assuming the manufacturer's claims are true, what is the probability that it finds a mean lifetime of less than 52 months?

SOLUTION

- By Central Limit Theorem, $\bar{X}$ is approximately normal

$$\mu = 54, \quad \sigma = 6$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{50}} = 0.85$$

- To find $P(\bar{X} < 52)$, we need to convert to z-scores:

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{52 - 54}{0.85} \approx -2.35$$

- Hence, use z-table to find $P(Z < -2.35)$
- Hence, the probability of this happening is 0.0094. We have (1- 0.0094)x100%= 99.06% certain that this will not happen.

EXAMPLE

- Radiation of microwave ovens has normal distribution with standard deviation = 0.6.
- A sample of 25 microwave ovens produced $\bar{X} = 0.11$
- Determine a 95% confidence interval for the mean radiation.

SOLUTION

The population is normal, and the observed value $\bar{X} = 0.11$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.6}{\sqrt{25}} = 0.12$$

$$P\left[0.11 - 1.96 \times \frac{0.6}{\sqrt{25}} < \mu < 0.11 + 1.96 \times \frac{0.6}{\sqrt{25}}\right] = 0.95$$

$$P[-0.1252 < \mu < 0.3452] = 0.95$$

(−.1252, 0.3452) is a 95% confidence interval for μ.

EX: 95% CONFIDENCE INTERVAL

- Suppose a sample of 100 students with mean SAT score of 1020, standard deviation of 200
  - Find the 95% Confidence Interval for the population mean?
- We know that:
  - 1. The sampling distribution is normally distributed.
  - 2. Therefore 95% of samples will yield a mean estimate within 2 standard deviations ($z = \pm 1.96$) of the population mean (μ).
- Thus, 95% of the estimates are within two "standard errors" of $\bar{X}$

SOLUTION

- 95% Confidence Interval

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$\boxed{\pm 1.96}$

$$\bar{X} \pm 2(\sigma_{\bar{X}}) = 1020 \pm (2)\left(\frac{200}{\sqrt{100}}\right)$$

$$= 1020 \pm 2 \times 20$$

$$= 1020 \pm 40$$

95% CI : (980, 1060)

CONFIDENCE INTERVAL FOR MEAN

- Similarly,

$$68\%\ \text{CI} : \bar{X} \pm 1\left(\frac{\sigma}{\sqrt{n}}\right) = 1020 \pm 20$$

$$99\%\ \text{CI} : \bar{X} \pm 3\left(\frac{\sigma}{\sqrt{n}}\right) = 1020 \pm 3 \times 20$$

- Q : Which one is a larger range?
- The larger that range, the more likely that the true mean will fall in it

EXAMPLE

- A system administrator monitored the execution time required to process applications.
- A random sample of 40 applications gave a sample mean time of 3.8 minutes and a standard deviation of 1.2 minutes.
- Construct a 90% confidence interval for the mean time to process, μ.

SOLUTION

- n = 40, $\bar{X} = 3.8$, σ = 1.2, 1 - α = 0.90

$$P\left[\bar{X} - Z_{\frac{\alpha}{2}=0.05} \times \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}=0.05} \times \frac{\sigma}{\sqrt{n}}\right] = 0.90$$

$$P\left[3.8 - 1.65 \times \frac{1.2}{\sqrt{40}} < \mu < 3.8 + 1.65 \times \frac{1.2}{\sqrt{40}}\right] = 0.90$$

$$P[3.487 < \mu < 4.113] = 0.90$$

- 90% CI : (3.487, 4.113)

NORMAL DATA EXAMPLE

Using $(X_1, \ldots, X_{10}) = (1874, 1642, 1603, 1931, 2103, 2068, 1948, 1798, 2364, 1918)$,

find point estimates of (μ, σ^2)

$$\mu = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X} = 1924.9 \qquad \sigma^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n} = 44797.09$$

$$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

unbiased point estimator

and find a 95% CI

$$\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}} = 1924.9 \pm 1.96\frac{\sqrt{44797.09}}{\sqrt{10}}$$

$$= 1924 \pm 131.2$$

$$= (1793.7, 2056.1)$$

In the example, n = 10, which gives d.f. = 10 - 1 = 9,

Try: Sample variance: $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1} = 49774.54$

So., we choose the value 2.262 in the t-distribution table to replace z=1.96, then a 95% CI becomes

$$1924.9 \pm 2.262 \frac{\sqrt{49774.54}}{\sqrt{10}} = 1924.9 \pm 159.587$$
$$= (1765.313, 2085.486)$$

Note that this interval is wider than the interval using 1.96.

In fact, a CI based on a t-table is always wider than its equivalent using the Z. The idea is, for small sample sizes, S^2 may not be a very accurate estimate of σ^2 and a wider interval introduce more uncertainty.

- For comparison, we replace 1.96 with a value from a t-table. The values from a t-table depend on a quantity called the degree of freedom (d.f.), which is defined as d.f. = $n - 1$. Values for selected d.f.'s are given below:

df = $n - 1$	**6**	**7**	**8**	**9**	**10**	**20**	**120**	**>120**
value	2.447	2.365	2.306	2.262	2.228	2.086	1.98	1.96

IMPROVING UPON A CI: REDUCING THE MARGIN OF ERROR

- The margin of error for a 95% CI is $1.96\frac{\sigma}{\sqrt{n}}$, which depends on σ and n
- $\sigma^2 = \mathrm{var}(X)$ measures the variation of X in the population, which is beyond our control. However, $\boldsymbol{n}$ is the sample size, which is under our control

- n can be increased to reduce the margin of error

Ex: What sample size m would reduce the margin of error by a factor of ½ ?

$$1.96\frac{\sigma}{\sqrt{m}} = \frac{1}{2}\left(1.96\frac{\sigma}{\sqrt{n}}\right)$$

$$\frac{1}{\sqrt{m}} = \frac{1}{2}\left(\frac{1}{\sqrt{n}}\right)$$

$$\frac{1}{m} = \frac{1}{4}\left(\frac{1}{n}\right) \quad \rightarrow \quad m = 4n$$

- To reduce the margin of error by a factor of 1/k, the sample size needs to be increased by k^2 times

A CONFIDENCE INTERVAL TO ESTIMATE A POPULATION PROPORTION

- A population proportion by p
- The point estimate of the population proportion is the **sample proportion** $\hat{p}$

Interval Estimate = $\hat{p} \pm$ **Margin of Error**

= $\hat{p} \pm Z_{\alpha/2}$ **(standard error)**

$$Standard\ error = S_e = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

EXAMPLE

- Finding the 95% Confidence Interval for a Population Proportion

95% CI = [point estimate ± margin of error]

$= \hat{p} \pm 1.96(standard\ error)$

EXAMPLE

Each year the 3rd year students may (S) or may not (F) choose to study Probability and Statistics.

To estimate the fraction who do study Probability and Statistics a sample of 1000 students was chosen from the past 5 years and 637 had chosen Probability and Statistics.

Construct a 95% confidence interval for proportion.

$$\hat{p} = \frac{\text{number of S student in sample}}{\text{Sample size}} = \frac{637}{1000} = 0.637$$

$$S_e = \sqrt{\frac{0.637(1 - 0.637)}{1000}} = 0.015$$

95% confidence interval = $0.637 \pm 1.96 \times 0.015$

= (0.607, 0.667)

EXAMPLE

A manufacturer of CD players uses a set of comprehensive tests to access the electrical function of its product. All CD players must pass all tests prior to being sold.

A random sample of 500 CD players resulted in 15 failing one or more tests. Find a 90% confidence interval for the proportion of CD players from the population that pass all test.

SOLUTION

N = 500, $\hat{p} = 0.97$, $Z_{0.05}$ = 1.645

$$0.97 \pm (1.645)\sqrt{\frac{(0.97)(0.03)}{500}} = 0.97 \pm 0.013$$

0.957 < *p* < 0.983

END CHAPTER 6